

# **Telnet and SSH Setup**

**BY**

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## Introduction

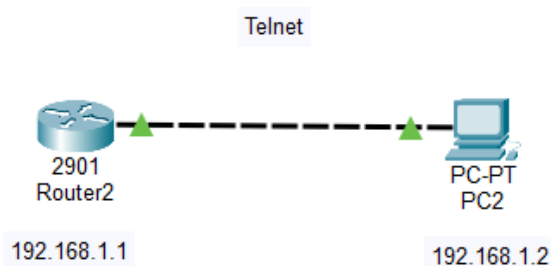
Remote access is an important part of computer networking and system administration, as it allows administrators to configure and manage network devices without being physically present at the device. Two commonly used protocols for remote access are Telnet and Secure Shell (SSH). Telnet provides a simple command-line interface for accessing remote devices, while SSH provides a more secure method by encrypting the communication between the client and the server.

This report demonstrates the configuration and use of Telnet and SSH in Cisco Packet Tracer. A PC and router are connected and configured with appropriate IP addresses, followed by the required settings for remote access. The report explains the basic configuration commands, authentication methods, and the process of accessing the router remotely from the PC. It also highlights the security difference between Telnet and SSH, particularly the use of encryption and authentication in SSH.

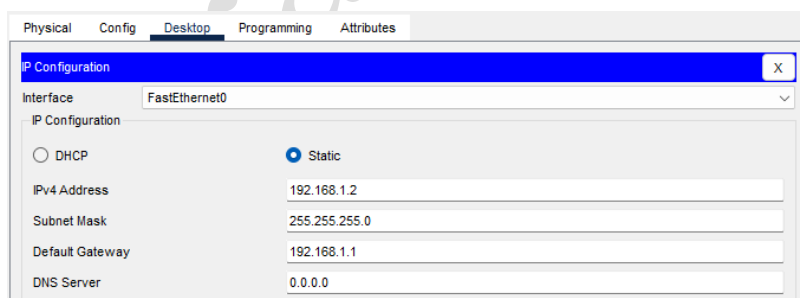
## Telnet Setup

Telnet is a classic network protocol that allows you to connect to and manage a remote computer or device over a network using a text-based, command-line interface. Operating on a client-server model, your local machine runs a Telnet client to send commands directly to a remote machine running a Telnet server, typically established over a Transmission Control Protocol (TCP) connection via port 23. This setup creates a virtual terminal, making it feel as though you are typing directly into the remote system.

To setup a telnet protocol in Cisco Packet Tracer. We establish a router and a PC, connect them and provide them IP accordingly.



First open the PC and assign the IP in IP config:



Then open router and set it up for IP and password:

```

Physical  Config  CLI  Attributes
IOS Command Line Interface

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#enable secret pass
Router(config)#interface gig
Router(config)#interface gigabitEthernet 0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#line vty 0 4
Router(config-line)#password pass123
Router(config-line)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#wr
Building configuration...
[OK]
Router#exit

```

The first password which we set using enable secret command is actually the password for router i.e. to access the router. While the other password is setup for telnet connection when accessing the protocol for connecting the PC to router port. The command line vty is for how many ports you want to establish for the tenet protocol. To do so open the command prompt on PC and connect to telnet through router by entering telnet and the IP for router as follows:

```

Physical  Config  Desktop  Programming  Attributes
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>telnet 192.168.1.1
Trying 192.168.1.1 ...Open

User Access Verification

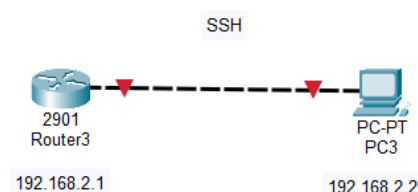
Password:
Router>enable
Password:
Router#

```

## SSH Setup

SSH, or Secure Shell, is a cryptographic network protocol that establishes an encrypted connection between two computers to allow for secure remote access, system administration, and data transfer. It operates on a client-server architecture—typically over TCP port 22—ensuring that all commands, credentials, and data transmitted across an unsecured network (like the internet) remain private and unaltered.

To setup a SSH protocol in Cisco Packet Tracer. We establish a router and a PC, connect them and provide them IP accordingly.



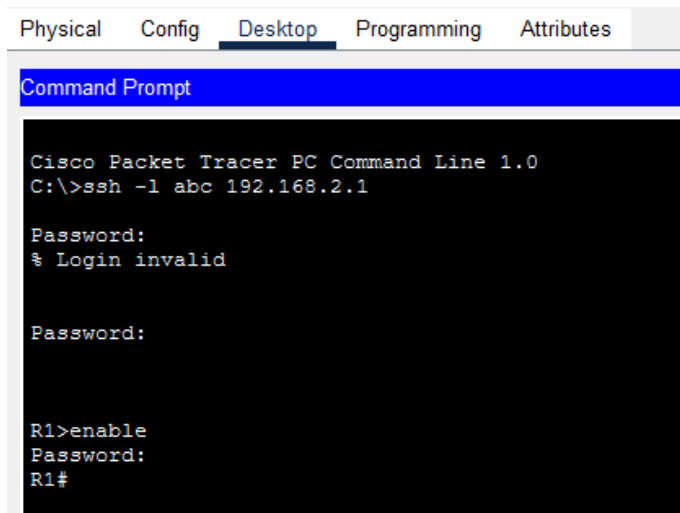
First open the PC and assign the IP in IP config:

| Physical                   | Config | Desktop                                 | Programming | Attributes |
|----------------------------|--------|-----------------------------------------|-------------|------------|
| <b>IP Configuration</b>    |        |                                         |             |            |
| Interface                  |        | FastEthernet0                           |             |            |
| IP Configuration           |        |                                         |             |            |
| <input type="radio"/> DHCP |        | <input checked="" type="radio"/> Static |             |            |
| IPv4 Address               |        | 192.168.2.2                             |             |            |
| Subnet Mask                |        | 255.255.255.0                           |             |            |
| Default Gateway            |        | 192.168.2.1                             |             |            |
| DNS Server                 |        | 0.0.0.0                                 |             |            |

Then open CLI on router and enter following commands. The commands which are new are after we have setup IP address for Ethernet 0/0. The command after that is username and password this command is used for SSH protocol used for accessing the router protocol through PC. After that we assign domain name for protocol to be able to generate the RSA key for encryption, which we do in the next command. After we enter the command it asks us for encryption key size, we can enter the default value i.e. 512 or we can go for more. The size entered is 2048 which will automatically ensure that the SSH version 2 is enabled, but we can also have small size key and version 2 SSH to do that we have to enter another command i.e. **ip ssh version 2**. Why use ver. 2, because its more secure that ver. 1. Then we enable required number of ports and transport them to SSH. Feed them to access for local devices.

| Physical                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Config | CLI | Attributes |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|-----|------------|
| IOS Command Line Interface                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |        |     |            |
| <pre>Router&gt;enable Router#configure terminal Enter configuration commands, one per line. End with CNTL/Z. Router(config)#hostname R1 R1(config)#enable secret pass1234 R1(config)#interface gig R1(config)#interface gigabitEthernet 0/0 R1(config-if)#ip address 192.168.2.1 255.255.255.0 R1(config-if)#no shutdown  R1(config-if)# %LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up  %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up  R1(config-if)#exit R1(config)#username abc password abc123 R1(config)#ip domain-name xyz R1(config)#crypto key generate rsa The name for the keys will be: R1.xyz Choose the size of the key modulus in the range of 360 to 4096 for your General Purpose Keys. Choosing a key modulus greater than 512 may take a few minutes.  How many bits in the modulus [512]: 2048 % Generating 2048 bit RSA keys, keys will be non-exportable...[OK]  R1(config)# *Mar 1 0:10:28.686: %SSH-5-ENABLED: SSH 1.99 has been enabled R1(config)#line vty 0 9 R1(config-line)#transport input ssh R1(config-line)#login local R1(config-line)#exit</pre> |        |     |            |

After all the setup is done access the port through the PC command prompt using telnet command:



```
Physical  Config  Desktop  Programming  Attributes

Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ssh -l abc 192.168.2.1

Password:
% Login invalid

Password:

R1>enable
Password:
R1#
```

Here -l means username followed by the username we assigned in router followed by routers' IP address. Then enter the password which is mentioned after username.

## Conclusion

The configuration of Telnet and SSH in Cisco Packet Tracer demonstrates how network devices can be remotely accessed and managed through a command-line interface. Telnet provides a straightforward method for remote access, but it does not encrypt transmitted information, making it unsuitable for secure communication over untrusted networks.

SSH provides a more secure alternative by encrypting the communication and using authentication to protect remote access. Through this practical setup, the process of configuring IP addresses, user credentials, VTY lines, and SSH keys was understood. Overall, the exercise demonstrates the importance of using secure remote-access protocols such as SSH when managing network devices in real-world networking and cybersecurity environments.